Homework 4
Due: Mar 13, 2025
Bonus problem (5 points)

Capillary pressure measurements are often unavailable far from the wellbore in the subsurface where no rock sample has been recovered. As a result, operators approximate it from its scaled value versus wetting phase saturation as follows:

$$J(S_w) = \frac{P_c(S_w)}{\gamma \cos(\theta)} \sqrt{\frac{k}{\varphi}} \tag{1}$$

where J is the scaled capillary pressure (unitless), often referred to as the *J* function, S_w is the wetting phase saturation (in fraction or %), P_c is the capillary pressure (in Pascal), γ is the interfacial tension (in N/m), θ is the contact angle (in degrees), k is permeability (in m^2), and φ is porosity (in fraction).

The *J* function is based on the notion that it converges for different samples at a given wetting phase saturation. Fig. 1 shows that the *J* function scales well at high wetting phase saturations (S_w>60%), but it performs poorly over a wide range of wetting phase saturations (S_w<50%). The significant scatter indicates that the *J* function cannot capture the underlying trend.

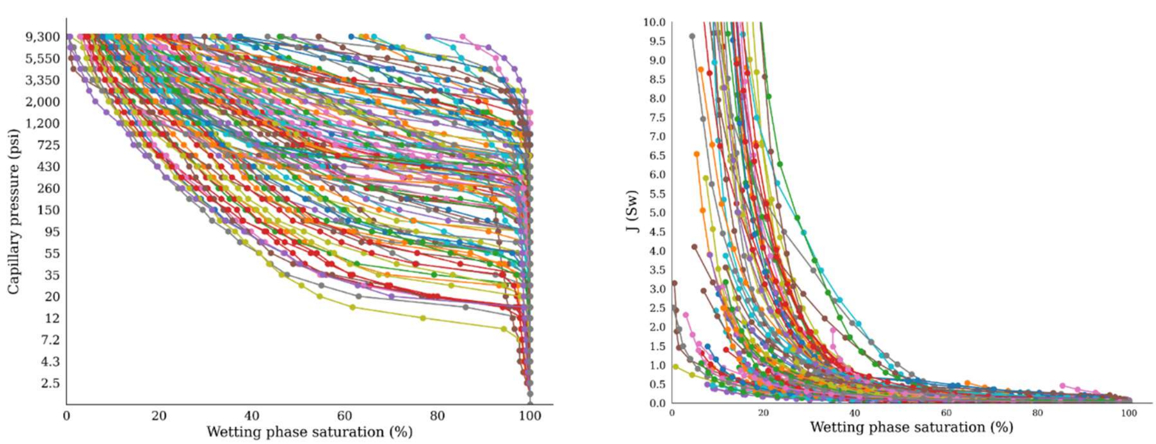

Figure 1. The capillary pressure measurements and *J* function representation of tight gas sandstones.

Three independent variables control the capillary pressure of a rock sample: permeability, porosity, and wetting phase saturation, $Pc=f(k,\varphi, S_w)$. The *J* function reduces them while keeping the dependency on the wetting phase saturation, $J=J(S_w)$, because it is possible to estimate S_w even if rock samples are not recovered. The *J* function was obtained using dimensional analysis, which differs from the dimensionality reduction.

The capillary pressure measurements of 106 rock samples were obtained using mercury (γ =484 dyne/cm, θ=140^o). The capillary pressure, permeability, and porosity measurements are posted on Canvas (pc_hw4, k_phi). Use Principal Component Analysis (PCA) or any other projection method you prefer to improve the representation with wetting phase saturation. To evaluate the performance, determine the ratio of the maximum to the minimum value of your projected results versus wetting phase saturation. Compare the ratio with that of the *J* function.

Hint: Determine the *J* function after converting the units to analyze its behavior. Note that each rock sample has one porosity, one permeability, and 36 capillary pressure measurements.